# Predicting Spotify Song Popularity from Audio Features and Lyrics

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# Motivation and main research question

## What caught our attention

We use Spotify a lot, and sometimes songs with very simple lyrics suddenly go viral on Instagram Reels. This made us curious: is popularity more related to how a song sounds, or what its lyrics say?

## Baby Shark question

Baby Shark has a catchy melody and repetitive lyrics. But which part makes it so memorable — the melody, the lyrics, or both?

## Why it matters for new songs

New songs often have little or no listening history. If lyrics and audio features can provide useful information, recommendation systems could use them to better understand new songs.

## Why it matters for artists

Artists and record labels may also benefit from knowing which song features are associated with popularity, such as lyric length or repetition.

**Main research question**

**Which carries more information about Spotify song popularity, audio features or lyrics, and how large is the gap?**



# Sub-questions and when we answer them

| | | |
|---|---|---|
| **SQ1** | Which audio features are most strongly associated with popularity? | Midterm |
| **SQ2** | Which lyric features (length, repetition, sentiment, keywords) are associated with popularity? | Midterm |
| **SQ3** | Does an audio-only or a lyrics-only model predict popularity better, and by how much? Does combining them help? | Final |
| **SQ4** | Do popularity drivers differ by genre? Which genres are hardest to predict? | Final |
| **SQ5** | How does popularity differ across release eras (2009–2025)? | Final |

- **Midterm focus:** SQ1 and SQ2, plus a reproducible cleaning pipeline, a shared split and an official baseline.
- **Final:** SQ3 answers the main research question; SQ4 and SQ5 add genre and era analysis.
- **Hypothesis H1:** Both the audio-only and lyrics-only models will outperform the baseline, but we expect the audio-only model to perform better than the lyrics-only model.

# Related work: audio, lyrics, or both

| Approach | Studies | What they report |
|---|---|---|
| **Audio only** | Jiang (2024); Araujo (2020); Herremans et al. (2014) | Danceability, energy and loudness matter most; the most and least popular songs are hardest to predict. |
| **Lyrics only** | Singhi & Brown (2014) | 31 rhyme, syllable and meter features can separate hits from non-hits. |
| **Both** | Dhanaraj & Logan (2005); Zhao et al. (2023); Prevedello et al. (2024) | Lyrics slightly more useful than audio, but no big gain when combined (2005); lyric embeddings improve early predictions (2024). |

# Data: from 550,622 rows to 171,510 songs

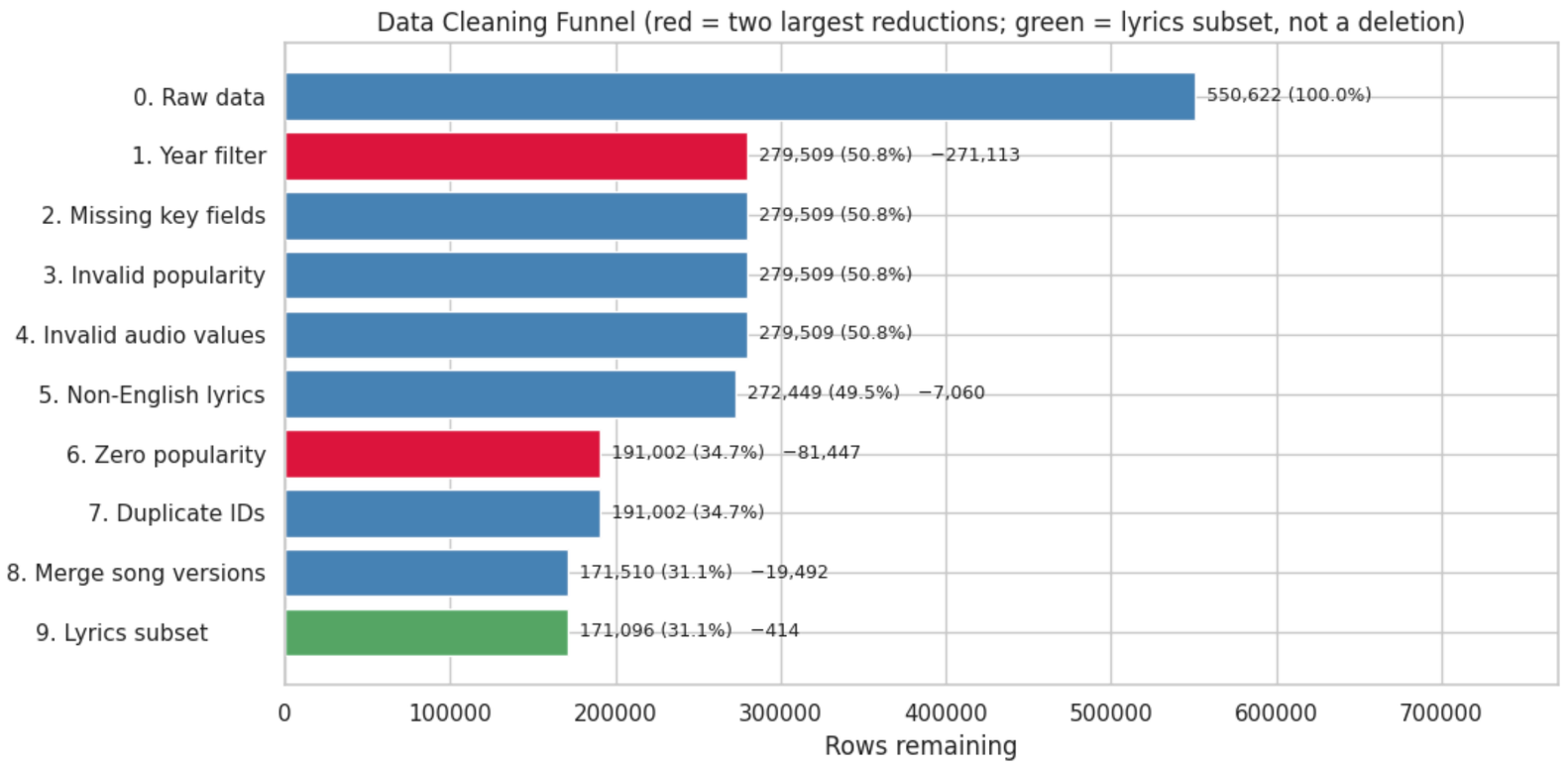



*Figure 1. Cleaning funnel (red = two largest reductions; green = lyrics subset, not a deletion).*

- **Source:** "550K Spotify Songs: Audio, Lyrics & Genres" on Kaggle (Serkan Tüysüz). Lyrics from LRCLIB; audio features from Spotify.

## 0.Raw Data

We started with 550,622 rows containing Spotify track information, audio features, and lyrics.

## 1.Year Filter

We kept songs from 2009 to 2025 to focus on more recent music.

## 2.Missing Key Fields

We checked for missing essential information, but no rows needed to be removed.

## 3.Invalid Popularity

We checked for invalid popularity scores, and all values were valid.

## 4.Invalid Audio Values

We checked audio features for invalid values, and no rows needed to be removed.

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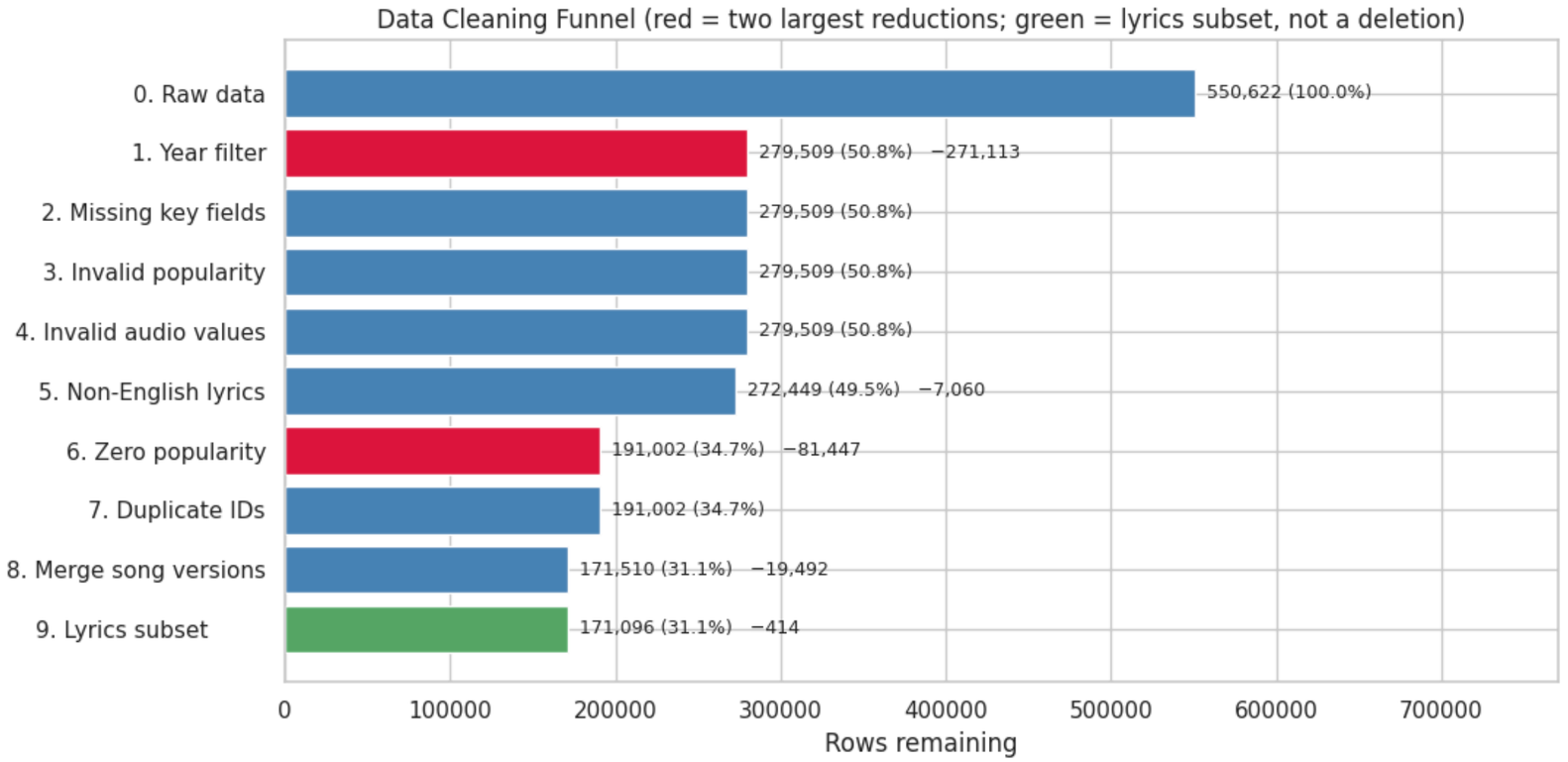



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## 5.Non-English Lyrics

We removed 7,060 rows with non-English lyrics to keep the language analysis consistent.

## 6. Zero Popularity

We excluded 81,447 rows with popularity scores of zero because their meaning was unclear.

## 7. Duplicate IDs

We checked for duplicate track IDs to avoid counting identical records more than once.

## 8. Merge Song Versions

We removed 19,492 extra rows by combining different versions of the same song into one record.

## 9. Lyrics Subset

We identified 171,096 songs with usable English lyrics for lyrics-related analysis, while keeping 171,510 songs for audio analysis.

# Data challenges: versions, zeros and shared artists

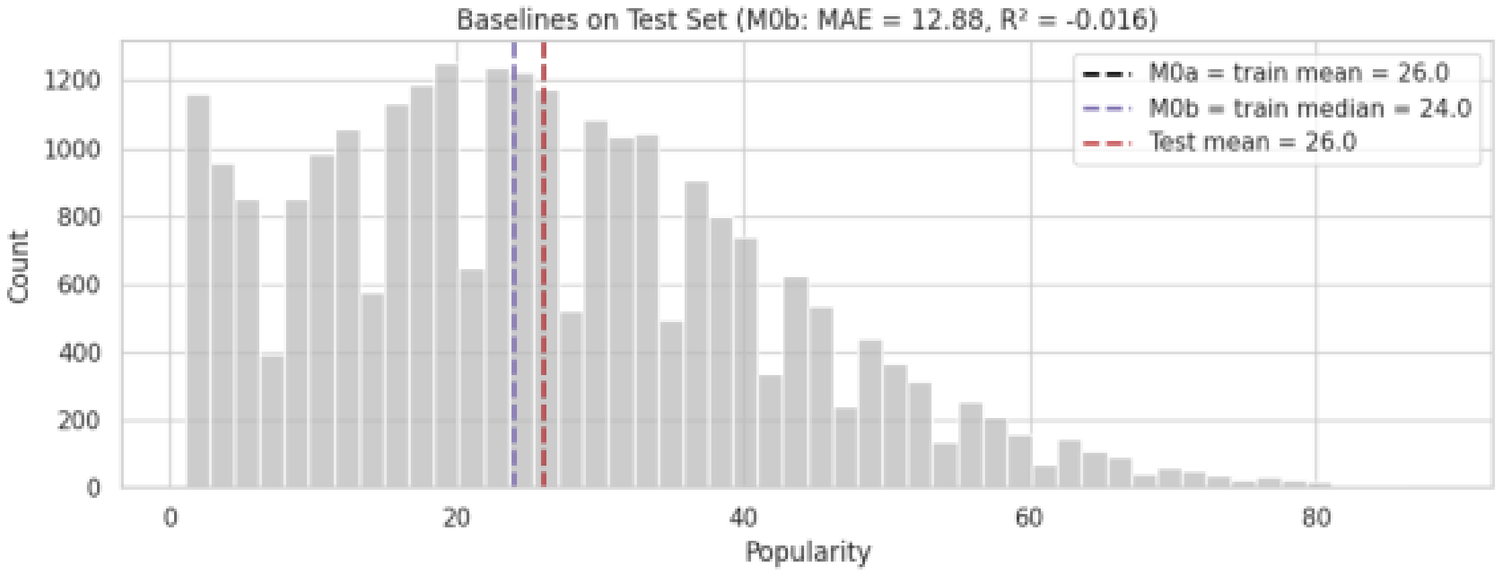



*Figure 2. Popularity on the test set; dashed lines show the baseline guesses (mean 26, median 24).*

- 171,096 songs in the main analysis
- Mean popularity: 26.0
- Median popularity: 24.0
- Popularity is right-skewed, so the median is lower than the mean

# Preliminary result 1: longer, more repetitive lyrics do slightly better

- **How to read the charts:** The left chart shows how each lyric feature is related to popularity (Spearman ρ). The right chart compares the top 10% most popular songs with the rest (Cohen's d).
- **Lyric length:** Word count has the strongest positive correlation with popularity (ρ = +0.28). Hit songs also tend to have longer lyrics (Cohen's d = 0.68).
- **Repetition:** Unique-word ratio has a negative correlation (ρ = −0.24), suggesting that songs with more repetitive lyrics tend to be slightly more popular.
- **Sentiment:** Positive and negative word ratios show very weak correlations (around ±0.05), suggesting that emotional tone provides little information on its own.

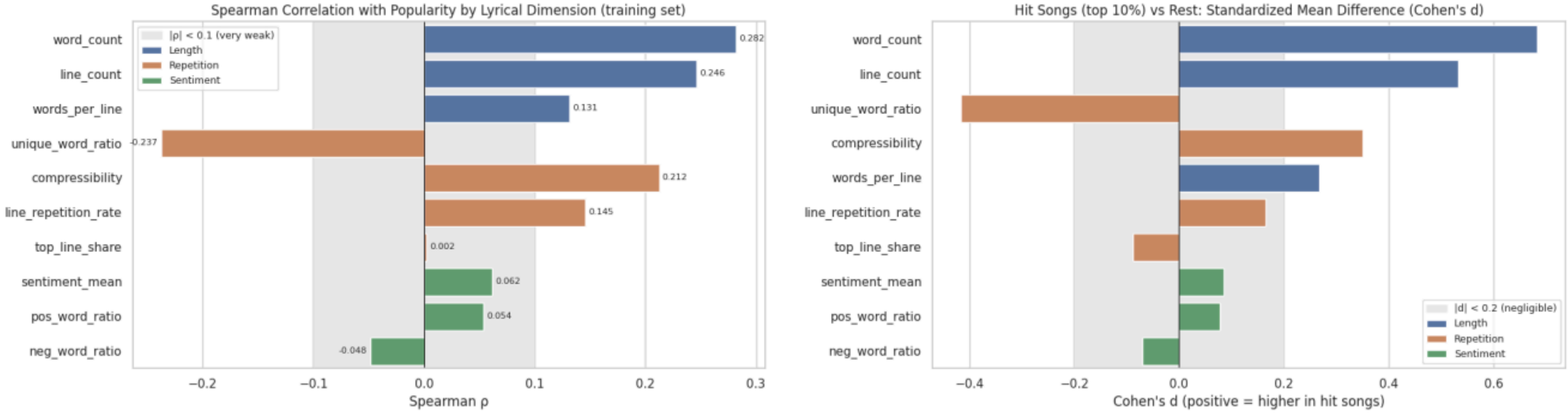



*Figure 3. (left) Spearman correlation of lyric features with popularity; (right) top-10% songs vs. the rest (Cohen's d).*

# Preliminary result 2

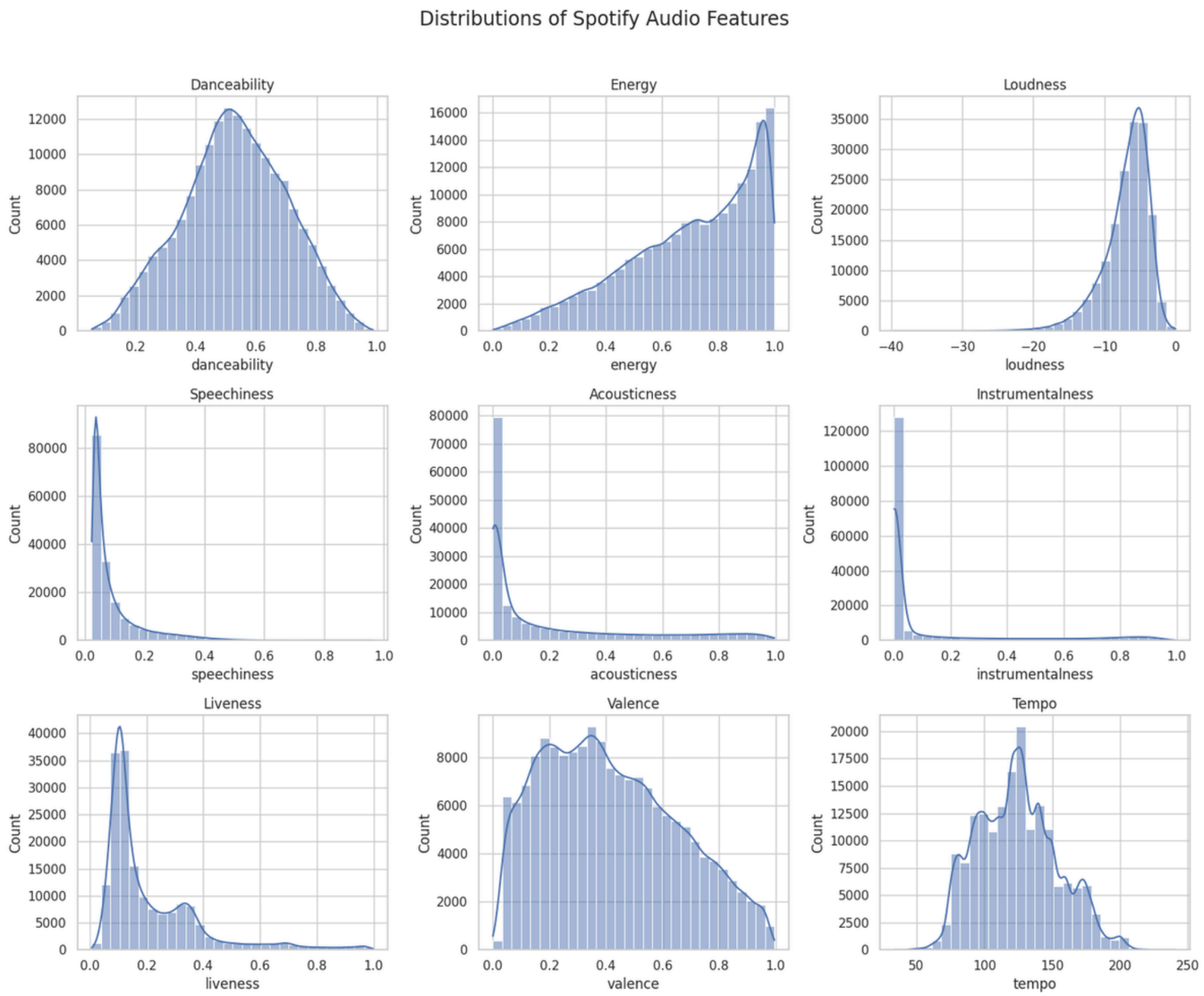



*Figure 4. Distributions of the nine Spotify audio features used as predictors.*

- **Danceability:** How suitable a track is for dancing
- **Energy:** How intense a track sounds
- **Loudness:** The overall loudness of a track, measured in decibels (dB)

- **Speechiness:** How much spoken word a track contains
- **Acousticness:** How likely a track is to be acoustic
- **Instrumentalness:** How likely a track is to contain no vocals

- **Liveness:** Detects the presence of an audience
- **Valence:** How positive the music sounds
- **Tempo:** The speed of the beat, measured in beats per minute (BPM)

# Preliminary result 2

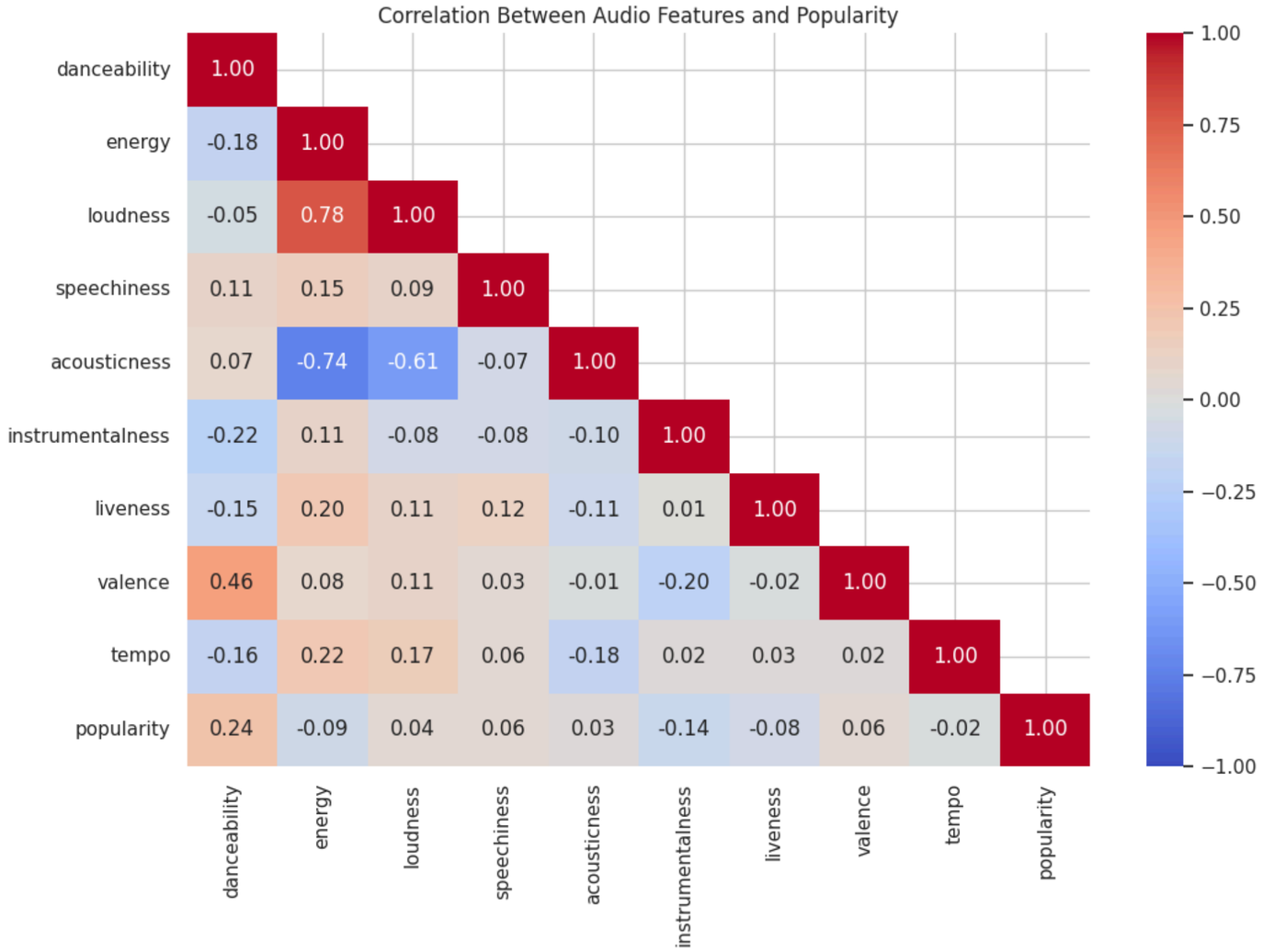



*Figure 5. Correlation matrix of the nine Spotify audio features and popularity; red indicates positive correlation and blue indicates negative correlation.*

**Energy & Loudness: +0.78**
Strong positive correlation.

**Danceability: +0.24**
Strongest positive correlation with popularity.

**Instrumentalness: −0.14**
Negative correlation with popularity.

**Tempo: −0.02**
Almost no linear correlation with popularity.

**Correlation does not mean causation.**

# Next: Model Selection

| Model | How It Works | Why We Use It |
| --- | --- | --- |
| **Baseline** | Predicts the same score for every song | A simple starting point |
| **Ridge Regression** | Learns patterns by combining song features | Simple and easier to understand |
| **Random Forest** | Combines many decision trees | Can learn more complex patterns |

## Our Goal

- Compare prediction accuracy.
- Find out whether more complex models perform better.

# Model Training and Evaluation

| Dataset | Purpose |
| --- | --- |
| **Training Set** | Teach the models |
| **Validation Set** | Choose model settings |
| **Test Set** | Evaluate final performance |

**How We Evaluate the Models**

- MAE (Mean Absolute Error): Measures the average prediction error.
- Lower MAE means more accurate predictions.

**Preventing Data Leakage**

- Keep different versions of the same song in the same data group.
- Do not use artist popularity or follower counts as input features.

# Thank you